THE TYPE OF A RATIONAL HODGE COUNTEREXAMPLE

BENJAMIN STANLEY FROHMAN

Abstract. The Clay Millennium statement of the Hodge conjecture is the *rational* sentence

$$\forall D\, \forall \gamma\, \exists z, \quad \mathrm{cl}(z) = \gamma,$$

with coefficients in $\mathbb{Q}$. This note records the two Lean 4 types of its negation, proves they are classically equivalent, and records that both types are uninhabited in the accompanying formalization (`BenFrohman/HodgeDisproof`). It does not produce a fourfold D_{bad} or a class γ_{bad}. The integral Hodge conjecture is a different sentence and is already false.

1. The Sentence

Let D range over smooth complex projective fourfolds and let γ range over rational Hodge classes of type $(2,2)$:

$$\gamma \in H^4(D, \mathbb{Q}) \cap H^{2,2}(D).$$

Let z range over rational algebraic cycles of codimension two and let cl be the cycle class map. The easy arrow—every such $\mathrm{cl}(z)$ is Hodge—is a theorem. The Clay problem is the converse, expressed by the universal sentence above.

Hodge's original integral sentence, with $\mathbb{Z}$ in place of $\mathbb{Q}$, is false: Atiyah–Hirzebruch (1961) produced torsion classes that are not algebraic; Kollár (1990) produced infinite-order integral classes that are not algebraic. Tensoring with $\mathbb{Q}$ kills torsion and identifies classes that differ by an integer multiple. Those terms do not inhabit the rational negation.

2. Two Types of Negation

In dependent type theory the constructive negation of a universal sentence is a function type, whereas a counterexample witness is a dependent pair (sigma/existential triple).

Lean Name	**Type**	**Inhabitant**
`RationalHodgeNegation`	$(\forall D\, \forall \gamma\, \exists z,\ \mathrm{cl}\, z = \gamma) \to \mathrm{False}$	a function
`RationalCounterexample`	$\exists D\, \exists \gamma,\ \forall z,\ \mathrm{cl}\, z = \gamma \to \mathrm{False}$	a triple $\langle D, \gamma, p\rangle$

Lean treats these as distinct types; the kernel will not accept a triple where it expects a function.

Constructive map (`counterexample_to_negation`). From a triple $\langle D_{\mathrm{bad}}, \gamma_{\mathrm{bad}}, p\rangle$ and a hypothetical proof h of the universal sentence, apply h at that host and class to obtain a cycle z, and feed z to p to obtain False. This direction requires no classical axiom.

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Copyright © 2026 Benjamin Stanley Frohman. All rights reserved. Licensed under the Apache License, Version 2.0 (`http://www.apache.org/licenses/LICENSE-2.0`). Companion Lean 4 repository: `https://github.com/BenFrohman/HodgeDisproof`. Zenodo DOI: 10.5281/zenodo.23627612.

Classical map (`negation_to_counterexample`). From a function sending every purported proof of the universal sentence to False, one does not constructively compute D_{bad}. Extracting the existential witness uses `Classical.not_forall`. The converse therefore depends on `Classical.choice`.

The kernel-checked lemmas in `Disproof/Type.lean` are `counterexample_to_negation`, `negation_to_counterexample`, and `equivalence`. Running `#print axioms equivalence` reports `[propext, Classical.choice, Quot.sound]`. That theorem relates two types without constructing an inhabitant of either.

3. The Miss Locus

Write $L(D)$ for the rational Hodge conjecture in codimension two on a polarized fourfold D, $\Delta_{\mathrm{Hdg}}(D)$ for the primitive rational $(2,2)$ classes, and

$$\Delta_{\mathrm{miss}}(D) = \Delta_{\mathrm{Hdg}}(D) \setminus \mathrm{im}(\mathrm{cl}_D).$$

By definition,

$$L(D) \iff \Delta_{\mathrm{miss}}(D) = \emptyset.$$

On a very general hypersurface fourfold of degree $d \geq 6$ in $\mathbb{P}^5$, the Lefschetz hyperplane theorem together with monodromy gives Noether–Lefschetz vanishing

$$H^4(X, \mathbb{Q}) \cap H^{2,2}(X) = \mathbb{Q}\langle \omega^2 \rangle,$$

hence $\Delta_{\mathrm{Hdg}}(D) = 0$ and $\Delta_{\mathrm{miss}}(D) = \emptyset$. The rational Hodge conjecture holds there because no primitive rational class remains to algebraize. A large complex Hodge number $h^{2,2}$ counts complex harmonic forms of type $(2,2)$, not rational Hodge classes.

A term of `RationalCounterexample` would require a *special* fourfold with $\Delta_{\mathrm{Hdg}} \neq 0$ and $\Delta_{\mathrm{miss}} \neq \emptyset$. That host is not constructed in this note.

4. Scope: What Is Asserted and What Is Not Asserted

(1) The integral Hodge conjecture is false (classical literature counterexamples cited below).
(2) The two rational-negation types `RationalHodgeNegation` and `RationalCounterexample` are classically equivalent in Lean 4 with zero `sorry`.
(3) The constructive direction `counterexample_to_negation` uses no classical axiom.
(4) Neither rational-negation type is inhabited in this repository, nor is the universal rational sentence itself.
(5) This note does not claim the Clay Millennium Prize, and does not claim that the rational Hodge conjecture is either true or false.

Appendix A. Verification certificate

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/HodgeDisproof`
- Verified commit:
  `a5e1af97dd70d041597542346cc3d37721dcd87c`
- Lean toolchain: `leanprover/lean4:v4.22.0`
- Mathlib: none (the development depends only on Lean core)
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/HodgeDisproof/actions/runs/41568063924`

- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:

  `https://github.com/BenFrohman/HodgeDisproof/archive/a5e1af97dd70d041597542346cc3d37721dcd87c.tar.gz`

  `a0ed8cf387e8706f80ea3c6b4789e32ddb13428cbe7783e7bbdc74f6a110c777`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 6 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`HodgeDisproof.missWitness_of_pairing`	`(no axioms)`
`HodgeDisproof.counterexample_to_negation`	`(no axioms)`
`HodgeDisproof.equivalence`	`[propext, Classical.choice, Quot.sound]`
`HodgeDisproof.negation_to_counterexample`	`[propext, Classical.choice, Quot.sound]`
`HodgeDisproof.not_exists_forall_not`	`(no axioms)`
`HodgeDisproof.not_forall_exists_not`	`[propext, Classical.choice, Quot.sound]`

References

[1] M. F. Atiyah and F. Hirzebruch, *Analytic cycles on complex manifolds*, Topology **1** (1961), 25–45.

[2] J. Kollár, *Trento examples*, in *Classification of irregular varieties* (Trento, 1990), Lecture Notes in Math., vol. 1515, Springer, 1992, pp. 134–139.

[3] C. Voisin, *Hodge Theory and Complex Algebraic Geometry I, II*, Cambridge Studies in Advanced Mathematics, Cambridge University Press, 2002.

[4] P. Deligne, *The Hodge Conjecture*, Clay Mathematics Institute Millennium Prize Problem description, 2000/2006.

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Independent Researcher, ORCID: 0009-0006-7068-3718
Email address: `frohmanbenjamin@gmail.com`